INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

ETHYLENE ABSORBER/SHELF-LIFE EXTENDER BLANKET, POTASSIUM PERMANGANATE, FOR FRESH FRUITS AND VEGETABLES

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers a porous fabric blanket containing ethylene-absorbing material. The blanket is intended to extend the shelf-life of fresh fruits and vegetables when it is used in their storage.

2. SALIENT CHARACTERISTICS

2.1 Absorbing materials. The absorbing material shall be activated alumina (aluminum oxide) coated or impregnated with potassium permanganate ($KMnO_4$) and shall have a maximum moisture content of 16 percent. The absorbing material shall be spherical or pelletized in shape and shall be porous for efficient removal of ethylene from the atmosphere. The absorbing material shall be bright purple in color and free of dust.

2.2 Construction. The blanket shall contain a range of 12 to 15 pounds of absorbing material. The fabric shall be a porous barrier such as non-woven polyester or similar material which allows passage of air, and is dense enough to prevent pellets or excessive particle fines from escaping to the atmosphere. The blanket shall be constructed to withstand tearing during normal placement in refrigerated walk-in boxes or shipping containers. The blanket shall be rectangular in shape and the surface area shall be in the range of 500 to 1200 square inches (in^2) per side. Eyelets to facilitate hanging shall be built into the outer borders of the blanket. Eyelets shall be placed so that the absorbing material will distribute evenly and not be compacted to one end when in a hanging position. A tag shall be affixed to the blanket and shall have the following statement: "AVOID CONTACT WITH WATER".

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Logistics Agency Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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2.3 Performance.

2.3.1 Moisture analysis. Analysis for moisture shall be made in accordance with the latest edition of the "Official Methods of Analysis of AOAC International":

Test	Source	Method
Moisture	Chapter: Cereal foods Section: Wheat flour	Air oven method

2.3.2 Potassium permanganate. The absorbing material shall have a minimum content of 4 percent $KMnO_4$ by weight to provide adequate capacity for ethylene removal. The determination for weight percentage of $KMnO_4$ shall be as follows:

a. Prepare a standard $KMnO_4$ solution by mixing 0.100 gram (g) $KMnO_4$ in 100 milliliters (mL) of water and then dilute it 20-fold (1 volume of $KMnO_4$ solution plus 19 volumes of water). This will produce a concentration of 0.005 percent by mass of $KMnO_4$.

b. Read the absorbency of the $KMnO_4$ solution on a spectrophotometer at 525 nanometers against water as the blank (A_{std}).

c. With a mortar and pestle, pulverize the absorbing material into a fine powder. Mix 1 g of the powder with 100 mL of water in a flask and stir gently for 10 minutes. Filter an aliquot of the solution.

d. Dilute the absorbing material solution 10-fold and read the absorbency at 525 nanometers (A_{sample}).

e. Determine percentage using the formula:

$$\text{Percent } KMnO_4 = (A_{sample}/A_{std}) \times (0.005 \text{ mass } \%) \times 1000$$

3. REGULATORY REQUIREMENTS

3.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Material safety data sheet (MSDS). A MSDS shall be furnished in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and 29 CFR 1910.1200, "Hazard Communication".

3.3 Hazard warning label. An Occupational Safety and Health Administration hazard warning label shall be furnished in accordance with 29 CFR 1910.1200.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, examinations, and tests.

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4.2 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

4.3 Market acceptability. The products offered must have been previously sold either to the government or on the commercial market.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 6.2(b)).

6. NOTES

6.1 Sources of documents.

6.1.1 CFR and FAR. Copies of CFR and FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of CFR documents may be obtained from http://www.gpoaccess.gov/ecfr/. Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/far/.

6.1.2 Federal standards. Copies of federal standards may be obtained from General Services Administration, Federal Supply Service, Specification Section, 470 East L'Enfant Plaza SW, Suite 8100, Washington, DC 20407. Electronic copies may be obtained from https://assist.daps.dla.mil/.

6.1.3 AOAC. Copies of the Official Methods of Analysis of AOAC International may be obtained from AOAC International, 481 North Fredrick Avenue, Suite 500, Gaithersburg, MD 20877-2417. Electronic copies may be obtained from http://www.aoac.org/.

6.2 Ordering data. The acquisition order should specify the following information:

a. CID document number and revision.
b. Preservation, packing, and marking (see 5.1)

6.3 Subject term (key word) listing.

Alumina
Aluminum oxide

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MILITARY INTERESTS:

Custodians:
Army - GL
Navy - MC
Air Force - 68
DLA - GS

Review Activities:
Army - MD1
Navy - SH
DLA - SS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FAS

Preparing Activity:
DLA - GS3

(Project 6850-2011-001)

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